QUONTECH

ASSET MANAGEMENT

Asset Management Challenge 2026

Round 1 – Strategic Asset Allocation

Portfolio D – Inflation Resilient (Equal Weight)

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1 Investment Policy

Investment Mandate and Objective. QuonTech manages a **USD 100 million** Strategic Asset Allocation designed to deliver competitive risk-adjusted returns across changing market conditions. Our investment philosophy combines participation in global economic growth with diversification across interest-rate, inflation and commodity-related risks. Performance is evaluated primarily through the **Sharpe Ratio**, subject to a target annualized volatility of **8%** and a permitted range of **7–9%**.

Rather than relying on a single macroeconomic scenario, the portfolio seeks balanced exposure to complementary return drivers, while maintaining transparent portfolio construction and disciplined risk oversight.

Strategic Allocation and Investment Universe. The selected strategy, *Portfolio D – Inflation Resilient*, allocates capital equally across nine liquid, USD-denominated ETFs. Equities represent approximately **44.44%** of capital, combining US and developed international markets with dedicated healthcare and energy exposure. Fixed income accounts for **33.33%**, emphasizing short-duration government and investment-grade corporate bonds alongside inflation-linked Treasuries. The remaining **22.22%** is invested in gold and diversified commodities.

This allocation is intended to capture equity-market upside while limiting sensitivity to nominal interest-rate shocks and introducing assets that may respond differently to inflationary and geopolitical developments. Equal weighting provides a transparent, reproducible allocation rule, reducing dependence on uncertain expected-return estimates.

Portfolio Holdings and Risk Contributions.

Table 1: Bloomberg implementation (16 October 2026 close) and estimated risk contributions

ETF	Last price (USD)	Shares	Invested (USD)	Actual weight	Risk contr.
IVV	766.67	14,492	11,110,581.64	11.1106%	14.59%
VEA	69.49	159,895	11,111,103.55	11.1111%	16.92%
XLV	176.71	62,877	11,110,994.67	11.1110%	12.65%
XLE	68.24	162,824	11,111,109.76	11.1111%	23.78%
TIP	105.54	105,278	11,111,040.12	11.1110%	5.52%
SHY	81.36	136,567	11,111,091.12	11.1111%	1.18%
IGSB	51.40	216,169	11,111,086.60	11.1111%	2.53%
GLD	382.01	29,085	11,110,760.85	11.1108%	11.28%
PDBC	20.11	552,516	11,111,096.76	11.1111%	11.55%
ETF total	–	**1,439,703**	**99,998,865.07**	**99.998865%**	**100.00%**
Cash (USD)	–	–	**1,134.93**	**0.001135%**	–
Portfolio total	–	–	**100,000,000.00**	**100.000000%**	–

Note: Bloomberg closing prices of 16 October 2026; whole-share rounding leaves USD 1,134.93 in non-interest-bearing cash. Risk contributions refer to the theoretical equal-weight SAA.

Risk Budget and Portfolio Constraints. Risk management is embedded in the investment mandate. Estimated ex ante annualized volatility is **8.69%**, within the prescribed 7–9% range. The one-month 95% parametric Value at Risk is **4.13%**, equivalent to approximately **USD 4.13 million**. We additionally adopt a **15% maximum acceptable drawdown** as an internal risk-tolerance threshold, against which cumulative portfolio losses will be monitored.

The portfolio is long-only and excludes individual stocks, short positions, leverage and inverse or leveraged ETFs. These restrictions preserve transparent exposure to the selected asset classes and avoid amplification of market losses through borrowing or derivative leverage. For implementable portfolio-construction methods, a **2% minimum allocation** ensures meaningful positions, while a **25% maximum** limits concentration. Equal Weight meets both by construction; Risk Parity is assessed separately without these bounds. Residual cash arising from whole-share implementation is held in USD without interest.

Portfolio Risk Profile. Equal capital allocation does not imply equal risk exposure. Equities account for approximately **67.94%** of estimated portfolio volatility despite representing 44.44% of capital, with the energy ETF (XLE) contributing the largest individual share of risk at **23.78%**. This concentration is an acknowledged feature of the strategy, assessed alongside portfolio volatility, drawdown and cross-asset correlations.

The resulting allocation establishes a clear strategic benchmark for subsequent investment decisions, combining a diversified economic rationale with explicit portfolio constraints and measurable risk limits.

2 Market Outlook

2.1 Macroeconomic Outlook

As of 17 October 2026, the investment outlook for the final quarter of the year is shaped by resilient but uneven growth, renewed inflation pressures and tighter monetary policy. The following observations refer to information published before the portfolio submission date; the scenarios below remain forward-looking judgments rather than forecasts with assigned probabilities.

Economic growth and labour markets. US labour-market data point to slower hiring rather than a sharp contraction: nonfarm payrolls rose by 29,000 in September and the unemployment rate stood at 4.2% (BLS, 2 October 2026). The ECB's September staff projections put euro-area real GDP growth at 0.9% for 2026 (ECB, 10 September 2026). This combination suggests uneven regional growth and supports evaluating both broad equity exposure and defensive sectors.

Inflation. US headline CPI increased by 3.7% year on year in September, with energy prices contributing materially to the monthly increase (BLS, 14 October 2026). Eurostat reported euro-area annual inflation of 3.8% in September, compared with 3.2% in August; energy was a major contributor (Eurostat, 16 October 2026). These releases reinforce the relevance of inflation risk for nominal bonds, equity valuations and real assets.

Monetary policy. On 16 September, the Federal Reserve raised the federal funds target range by 25 basis points to 3.75–4.00%. On 10 September, the ECB increased its three key policy rates by 25 basis points. The direction of future policy remains conditional on incoming inflation and activity data; the portfolio therefore should not depend on an immediate return to monetary easing.

Energy and geopolitical supply risk. The International Energy Agency's Oil Market Report of 14 October describes continuing disruptions to oil supply and refined-product markets, alongside a sharp increase in benchmark crude prices during September. This provides a factual basis for considering energy-related shocks, but does not establish that energy equities or commodity ETFs will necessarily appreciate over the competition horizon.

2.2 Asset-Class Views

Equities. Developed-market equities provide participation in economic growth and corporate earnings. However, elevated discount rates, earnings disappointments and concentrated sector performance could amplify downside risk. Healthcare may offer relatively defensive earnings characteristics, whereas energy equities can benefit from stronger energy prices but remain exposed to broad equity-market and demand risks. Geographic diversification can reduce dependence on a single market, although correlations may rise during stress.

Fixed income. Short-duration government bonds and investment-grade corporate bonds can provide income with lower interest-rate sensitivity than longer-maturity securities. Corporate bonds nevertheless retain credit-spread risk. Inflation-linked government bonds offer protection against unexpected inflation through principal indexation, but their market prices can still decline when real yields rise. Longer-duration nominal bonds may perform well under disinflation but remain more exposed to adverse rate shocks.

Gold and commodities. Gold may diversify equity and bond risk during periods of financial or geopolitical uncertainty, although its performance also depends on real yields and investor demand. Broad commodities offer exposure to physical-market supply and demand and may respond positively to certain inflationary shocks. Neither is a reliable hedge in every scenario,

and commodity-futures returns also depend on the structure of futures markets.

2.3 Risks to the Outlook and Portfolio Implications

Three plausible scenarios guide the assessment of strategic exposures over the competition horizon.

Base scenario – Resilient but uneven growth. Economic activity remains positive, while inflation and monetary-policy uncertainty persist. Broad equities may participate in growth, but returns could vary substantially by region and sector. Short-duration fixed income may provide relative stability, while real assets offer exposure to different economic drivers.

Upside scenario – Faster disinflation. Inflation moderates without a major deterioration in earnings, allowing financial conditions to improve. Equities and nominal bonds could benefit from lower discount rates. Energy and broad commodities may underperform if inflation expectations and demand-driven price pressures decline.

Downside scenario – Supply shock or growth slowdown. A renewed energy shock could pressure corporate margins and nominal bonds while supporting selected commodity and energy exposures. A demand-driven recession, however, could weaken both equities and cyclical commodities; gold and high-quality government bonds may provide diversification, depending on real-yield and liquidity conditions.

3 Investment Universe

The initial investment universe comprises twelve US-listed, USD-denominated exchange-traded funds (ETFs) across equities, fixed income and real assets. The objective at this stage was to identify a sufficiently broad set of liquid instruments with distinct economic exposures, before defining the candidate portfolios and evaluating allocation methodologies.

The screening framework considered four criteria: market accessibility and implementation practicality; availability of a consistent monthly price history; exposure to relevant macroeconomic risk factors; and the ability to combine instruments into alternative strategic universes.

Equities. IVV and VEA provide broad developed-market participation, while VWO adds emerging-market exposure. XLV and XLE allow targeted assessment of healthcare and energy, whose sensitivities to economic growth, inflation and commodity prices may differ from those of the wider equity market.

Fixed income. The five bond ETFs span nominal government duration (IEF and SHY), diversified investment-grade bonds (AGG), short-term corporate credit (IGSB), and inflation-linked Treasuries (TIP). This distinction is important because nominal yields, real yields and credit spreads can affect bond returns differently.

Real assets. GLD and PDBC introduce exposures driven partly by safe-haven demand, real interest rates and physical commodity markets. Their diversification potential is conditional on the nature of the economic shock; neither provides guaranteed inflation protection.

The twelve ETFs are not twelve independent sources of risk. In particular, IVV already includes healthcare and energy companies represented in XLV and XLE, while several bond ETFs share interest-rate exposure. The analysis therefore considers portfolio-level correlations and risk contributions rather than treating the number of holdings as a sufficient measure of diversification.

This initial universe is subsequently divided into four candidate portfolios, each expressing a different strategic balance between growth, defensive characteristics and inflation sensitivity.

Table 2: Initial investment universe and economic exposures

ETF	Asset class	Underlying exposure	Role in the analysis
IVV	Equity	US large-cap stocks (S&P 500)	Broad US market participation
VEA	Equity	Developed markets ex-US	Geographic diversification
VWO	Equity	Emerging markets	Additional geographic and growth exposure
XLV	Equity	US healthcare sector	Defensive-sector characteristics
XLE	Equity	US energy sector	Energy-price and inflation sensitivity
IEF	Fixed income	7–10 year US Treasuries	Intermediate nominal-duration exposure
AGG	Fixed income	US investment-grade aggregate bonds	Broad nominal bond exposure
SHY	Fixed income	1–3 year US Treasuries	Short-duration government exposure
IGSB	Fixed income	Short-term investment-grade corporates	Short-duration credit exposure
TIP	Fixed income	US Treasury inflation-protected securities	Inflation-linked bond exposure
GLD	Real assets	Physical gold	Potential financial-stress diversifier
PDBC	Real assets	Diversified commodity futures	Commodity and supply-shock exposure

The composition of those portfolios is presented in Section 4, before any out-of-sample performance comparison.

3.1 Historical Risk and Return Inputs

The allocation methodologies were calibrated using 60 monthly USD ETF returns from September 2020 to August 2025. Table 3 reports the historical annualized mean returns and volatilities of the twelve ETFs. Sample mean returns serve as expected-return estimates for Maximum Sharpe Ratio optimization rather than forward-looking forecasts. The annualized risk-free-rate input is 2.87%.

Table 3: Historical return and volatility inputs (September 2020–August 2025)

ETF	Mean return (annualized)	Volatility (annualized)
IVV	15.10%	16.00%
VEA	11.19%	16.46%
VWO	7.16%	14.51%
XLV	7.38%	14.39%
XLE	27.02%	29.75%
IEF	-2.05%	7.60%
AGG	-0.48%	6.43%
SHY	1.37%	2.03%
IGSB	2.21%	3.27%
TIP	1.24%	6.17%
GLD	11.89%	14.29%
PDBC	12.69%	15.54%

The correlation matrix provides additional information on diversification and overlapping risk

exposures. For example, IVV and XLV exhibit a correlation of 0.70, while XLE is nearly uncorrelated with IEF (-0.01) and moderately positively correlated with TIP (0.20). XLE and GLD exhibit a negative correlation of -0.15. These relationships inform covariance estimation and portfolio risk contributions, although historical correlations may change under market stress.

Figure 1 displays the complete in-sample correlation matrix for the twelve ETFs. The highest positive correlations are concentrated among overlapping equity and fixed-income exposures, whereas XLE and GLD exhibit a modest negative correlation. The underlying numerical matrix is also available in the accompanying Excel workbook (`STATS in-sample`).

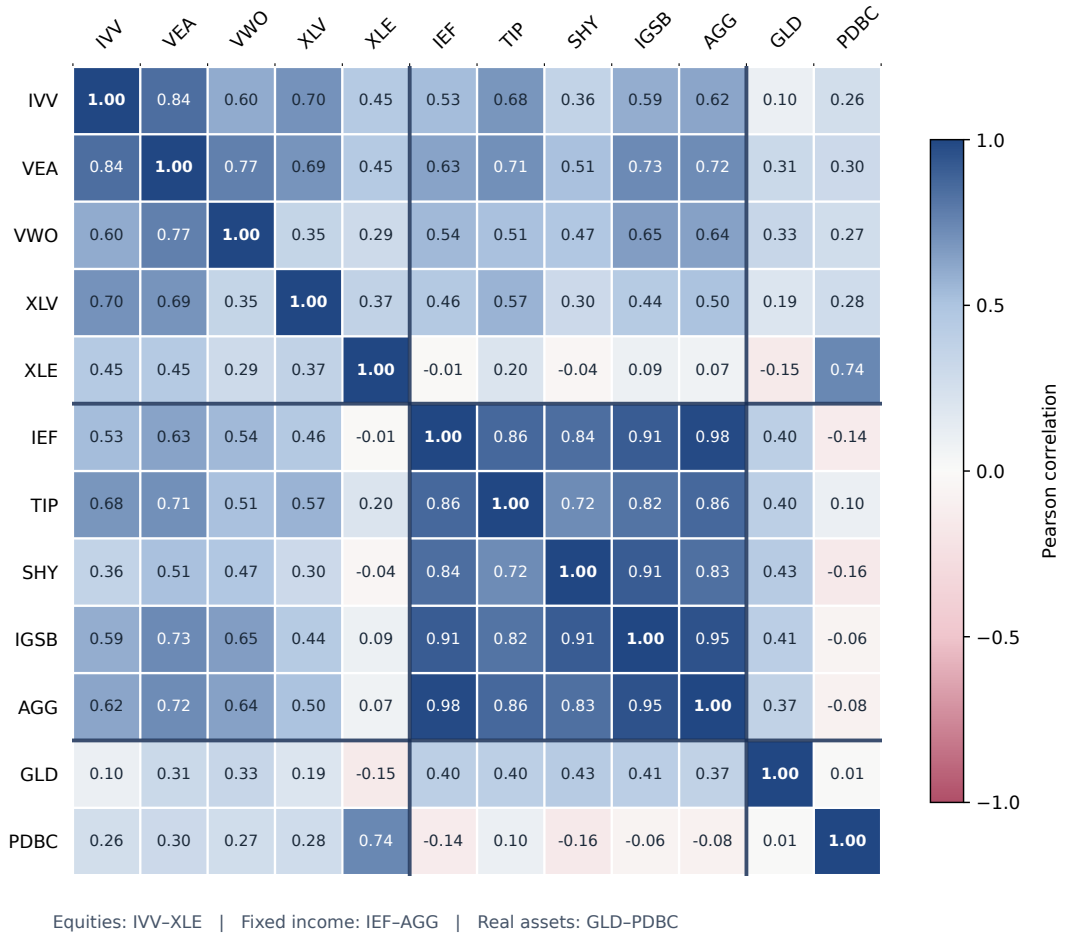


Figure 1: Pearson correlation matrix of monthly USD ETF returns (September 2020–August 2025; 60 observations). Blue indicates positive correlation, red negative correlation, and near-white values close to zero. Source: Bloomberg-derived returns, Quontech workbook, `STATS in-sample`.

Source: Bloomberg-derived monthly USD ETF returns, Quontech workbook, sheets `Window (in sample)` and `STATS in-sample`. Annualized mean returns are calculated by multiplying monthly sample means by 12, while annualized volatilities use the square root of 12. Correlations are monthly Pearson coefficients. The risk-free-rate input is based on the one-month US Treasury bill series.

4 Portfolio Construction

4.1 Four Candidate Universes

Four investment universes were constructed from the initial twelve ETFs, each representing a different strategic approach to growth, defensive positioning and inflation risk. Their composition was established before the out-of-sample evaluation.

Table 4: Candidate investment universes

Universe	Orientation	ETFs	Excluded from initial universe
A	Balanced	10	XLE, SHY
B	Defensive	9	VWO, XLE, PDBC
C	Growth	8	TIP, SHY, AGG, PDBC
D	Inflation Resilient	9	VWO, IEF, AGG

Universe A – Balanced. Retains developed- and emerging-market equities, a range of nominal and inflation-linked bonds, gold and broad commodities. It provides a diversified starting point without a pronounced sector tilt toward energy.

Universe B – Defensive. Excludes emerging-market equities, energy-sector equities and broad commodities. It emphasizes developed-market equities, healthcare, government bonds, investment-grade credit and gold, seeking to moderate exposure to cyclical and commodity-price shocks.

Universe C – Growth. Retains broad and emerging-market equities, healthcare and energy, alongside a narrower set of fixed-income instruments and gold. Its composition increases the relative importance of equity-market performance and economic growth.

Universe D – Inflation Resilient. Combines developed-market equities, healthcare and energy with inflation-linked bonds, short-duration Treasuries and corporate credit, gold and diversified commodities. Excluding IEF and AGG limits exposure to intermediate-duration nominal bonds, while retaining assets that may respond differently to inflationary or supply-side shocks.

These are alternative *investment universes*, not four sets of final portfolio weights. The weighting methodology is applied separately within each universe.

4.2 Allocation Methodologies

Seven weighting methodologies were considered for each universe, generating **28 portfolio–strategy combinations**. This design allows us to assess both the effect of asset selection and the effect of the allocation rule.

Maximum Sharpe Ratio. Chooses portfolio weights to maximize estimated excess return relative to volatility. Because it uses both historical expected returns and covariances, the resulting weights can be highly sensitive to estimation error.

Maximum Sharpe Ratio with Ledoit–Wolf shrinkage. Uses the same objective but replaces the sample covariance matrix with a Ledoit–Wolf shrinkage estimate, which pulls the sample matrix toward a scaled identity target. This regularizes covariance estimation; it does not remove uncertainty in expected returns.

Minimum Variance. Minimizes estimated portfolio variance without optimizing expected returns. The method can favor lower-volatility assets and may produce concentrated weights.

Minimum Variance with Ledoit–Wolf shrinkage. Applies the minimum-variance objective to a shrinkage covariance matrix to test whether covariance regularization improves weight stability.

Risk Parity. Seeks approximately equal contributions to portfolio volatility, accounting for individual asset risk and correlations. Risk Parity is evaluated without individual-weight bounds, allowing an unconstrained assessment of equal-risk allocation.

Inverse Volatility. Allocates capital inversely to estimated individual volatilities. Unlike Risk Parity, the basic rule does not explicitly incorporate cross-asset correlations.

Equal Weight. Assigns the same fraction of capital to each ETF in a given universe. It does not require estimates of expected returns or covariances to determine weights, making it a transparent benchmark for more estimation-intensive methods.

Weight bounds. For Maximum Sharpe, Minimum Variance (with and without shrinkage) and Inverse Volatility, each ETF included in a universe is constrained to between **2% and 25%** of capital. The 2% floor keeps every selected ETF in the portfolio and avoids corner solutions; the 25% cap limits concentration. Equal Weight satisfies both bounds by construction, while Risk Parity is left unconstrained.

All strategies are evaluated on a fully invested, long-only basis.

5 In-Sample Calibration

The four candidate universes were calibrated on the same 60 monthly USD return observations, from September 2020 to August 2025. This common estimation window allows differences in portfolio weights to be attributed to the investment universe and allocation methodology rather than to different historical samples. September 2025 was excluded as a separation period before the out-of-sample evaluation, which begins in October 2025.

5.1 Estimation Framework

Monthly ETF returns were used to estimate expected returns, individual volatilities and the covariance matrix, providing the inputs required by the different allocation methodologies.

Ledoit–Wolf covariance shrinkage was applied to Maximum Sharpe Ratio and Minimum Variance as a robustness check.

Excel was used to organize Bloomberg data, calculate historical returns and descriptive statistics, and develop the backtesting and risk analysis. Python was primarily used to implement the portfolio optimization methods and calculate the corresponding weights. Several performance and risk calculations were also reproduced in Python to verify the results and support reproducibility.

5.2 Calibration and Model Risk

All candidate portfolios were constructed using only information from the in-sample period. Long-only and full-investment conditions apply throughout; the 2–25% weight bounds apply to the optimized and inverse-volatility strategies, while Risk Parity is unconstrained.

For each candidate, the in-sample analysis considers risk-adjusted performance, volatility, drawdown and weight concentration. These statistics are diagnostic: optimizing against historical observations can produce an apparently attractive in-sample Sharpe Ratio even when the estimated weights are unstable. This concern is particularly relevant to Maximum

Sharpe Ratio optimization, because sample mean returns are difficult to estimate precisely from 60 observations.

Table 5: In-sample statistics of the strategies compared out of sample in Table 7 (September 2020–August 2025)

Portfolio and method	Sharpe	Volatility	Max. drawdown
D – Equal Weight	0.793	9.00%	–10.66%
D – Max Sharpe + LW	1.049	11.94%	–12.41%
D – Max Sharpe	1.049	11.63%	–12.26%
A – Max Sharpe	0.921	8.50%	–12.74%
C – Equal Weight	0.698	10.19%	–13.00%
C – Max Sharpe	1.025	11.20%	–11.12%
B – Max Sharpe + LW	0.708	9.71%	–16.92%

In sample, the Maximum Sharpe portfolios of Universes D and C rank highest (Sharpe ratios of 1.05 and 1.03), while D – Equal Weight ranks lower at 0.79. Out of sample the ranking reverses (Section 6), which is consistent with estimation error in optimized weights.

Portfolio selection is based on the subsequent out-of-sample comparison, using weights fixed during the in-sample calibration.

6 Out-of-Sample Backtest

6.1 Backtesting Protocol

To assess whether the portfolio construction methods remained effective beyond their estimation sample, we compared the 28 universe–strategy combinations using a common, strictly subsequent out-of-sample (OOS) window. Portfolio weights were estimated only from the in-sample observations; OOS returns were not used to re-estimate those weights.

Table 6: Historical estimation and evaluation windows

Stage	Period	Monthly observations
In-sample estimation	September 2020–August 2025	60
Separation period	September 2025	1 (excluded)
Out-of-sample evaluation	October 2025–September 2026	12

The excluded month provides a transparent separation between estimation and evaluation; it does not, by itself, eliminate selection bias or other backtesting biases. A twelve-month OOS window offers more observations than a three-month test matching the competition horizon, but remains short for estimating risk-adjusted performance precisely.

For a fixed-weight, monthly rebalanced portfolio, the OOS return in month t is $r_{p,t} = \sum_i w_i^{IS} r_{i,t}$. We compare annualized Sharpe ratios, realized volatility, maximum drawdown and cumulative wealth. The Sharpe ratio is the principal ranking metric, while the remaining measures provide complementary information about risk and return.

6.2 Comparison of Out-of-Sample Performance

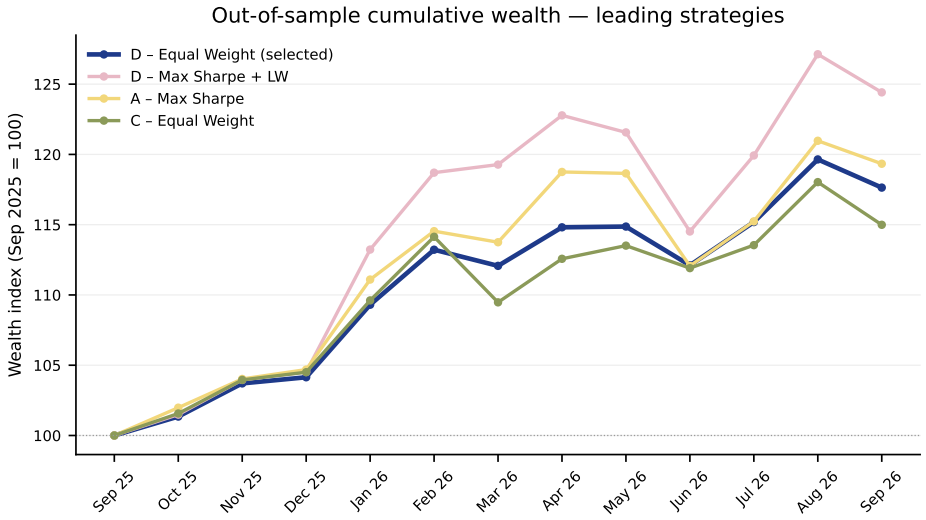


Figure 2: Out-of-sample cumulative wealth of D – Equal Weight and three leading alternatives, monthly rebalancing (September 2025 = 100). D – Max Sharpe + LW ends higher, but with markedly higher volatility and drawdowns. Source: Quontech workbook, OOS monthly ETF returns.

Table 7: Leading OOS results ranked by Sharpe ratio, and best strategy in Universe B

Portfolio and method	**Sharpe**	**Volatility**	**Max. drawdown**
D – Equal Weight	**1.598**	**8.08%**	**–2.40%**
D – Max Sharpe + LW	1.429	13.38%	–6.73%
D – Max Sharpe	1.420	13.15%	–6.61%
A – Max Sharpe[a]	1.320	11.08%	–5.63%
C – Equal Weight	1.124	9.54%	–4.09%
C – Max Sharpe[a]	1.113	12.06%	–4.74%
B – Max Sharpe + LW (best in B)	0.669	11.91%	–6.38%

[a] In Universes A and C, Max Sharpe + LW produces the same weights as Max Sharpe.

In the reported comparison, D – Equal Weight achieved the highest OOS Sharpe ratio (1.598), with realized volatility of 8.08% and a maximum drawdown of –2.40%. The strongest Maximum Sharpe alternatives had Sharpe ratios of approximately 1.32–1.43 but experienced higher realized volatility and deeper drawdowns. D – Max Sharpe + LW ended the period with higher cumulative wealth (Figure 2), but at roughly 13.4% volatility, well above the 7–9% mandate range. Thus, within this particular test, the additional estimation and optimization complexity did not translate into superior risk-adjusted performance.

This is not evidence that Equal Weight is universally preferable. Its outcome depends on the underlying ETF universe. In particular, Universe D includes XLE, whose favorable OOS performance materially supported its results. Universe B, which excludes XLE and PDBC, illustrates this: none of its seven strategies exceeded an OOS Sharpe ratio of 0.67, and its Equal Weight version reached only 0.43. Universe selection therefore matters as much as the weighting rule.

Out-of-Sample Value at Risk. Using the realized annualized OOS volatility of 8.08%, we estimate the one-month 95% parametric VaR under a normal approximation, without adjusting for the mean return:

$$\mathrm{VaR}^{OOS}_{95\%,\,1m} = 1.645\,\frac{8.08\%}{\sqrt{12}} \approx 3.84\%.$$

For the initial USD 100 million portfolio, this corresponds to approximately USD 3.84 million. The estimate is based on only twelve OOS monthly observations and is therefore subject to substantial estimation uncertainty. It differs from the 4.13% ex ante VaR reported in Section 9, which uses a separate 60-month risk-estimation window.

6.3 Implications for Portfolio Selection

The OOS comparison favors D – Equal Weight on the principal performance metric, while its realized risk profile is comparatively moderate. The final choice must nevertheless also be consistent with the ex ante volatility requirement and the forward-looking investment policy. Historical OOS volatility and ex ante risk-budget volatility are different statistics, calculated over different windows, and should not be treated as interchangeable.

6.4 Backtest Limitations

The interpretation of these results requires caution. First, twelve monthly returns provide an imprecise estimate of a strategy's long-run Sharpe ratio. Second, comparing 28 alternatives and subsequently selecting the best OOS performer introduces *selection bias*: the same evaluation window has influenced the final strategy choice. The selected portfolio's reported Sharpe ratio is therefore not a genuinely untouched estimate of prospective performance.

Additional limitations include the influence of favorable energy-sector returns, the exclusion of transaction costs and market impact, and potential differences between historical and future market conditions.

Discounted Sharpe Ratio. Given the limited twelve-month out-of-sample window, the selection bias arising from comparing 28 portfolio–strategy combinations, and the favorable contribution of energy-sector performance, we apply a conservative 50% haircut to the observed Sharpe Ratio of 1.598. This yields an adjusted Sharpe Ratio of approximately 0.80, which we adopt as a more prudent planning assumption for the competition. The discount reflects the team's judgment rather than a statistically estimated correction, and should not be interpreted as a reliable forecast of future performance.

7 Selected Portfolio: D – Equal Weight

7.1 Final Strategic Allocation

Quontech selected Portfolio D – Inflation Resilient, implemented with Equal Weight, based on its leading out-of-sample Sharpe Ratio, transparent allocation rule and compliance with the ex ante volatility requirement.

The strategic ETF targets comprise 44.44% equities, 33.33% fixed income and 22.22% real assets, before the negligible whole-share cash remainder. These are target capital allocations, not risk allocations. The portfolio deliberately excludes emerging-market equities and intermediate-duration nominal bond ETFs; the resulting exposures are evaluated in Section 8.

Table 8: Selected strategic allocation and investment rationale

ETF	Exposure	Weight	Role in the portfolio
IVV	US large-cap equities	11.11%	Broad participation in US equity growth
VEA	Developed ex-US equities	11.11%	International equity diversification
XLV	US healthcare equities	11.11%	Defensive sector tilt
XLE	US energy equities	11.11%	Energy-price sensitivity and inflation exposure
TIP	US inflation-linked Treasuries	11.11%	Inflation-linked bond exposure
SHY	Short-term US Treasuries	11.11%	Lower nominal-duration exposure
IGSB	Short-term IG corporate bonds	11.11%	Short-duration credit exposure
GLD	Gold	11.11%	Potential diversifier in financial stress
PDBC	Diversified commodities	11.11%	Exposure to physical commodity markets

7.2 Bloomberg Implementation: Closing Prices of 16 October 2026

The selected SAA was implemented using Bloomberg closing prices of 16 October 2026 and whole ETF shares. Minor differences between target and actual weights arise from share rounding, leaving USD 1,134.93 in non-interest-bearing cash. The implementation details are reported in Section 1.

7.3 Rationale for Equal Weight

Equal Weight does not rely on estimated expected returns or covariances to determine portfolio weights, reducing sensitivity to estimation error. This is particularly relevant with only 60 monthly calibration observations, as Maximum Sharpe optimization can generate concentrated positions from noisy expected-return estimates. Ledoit–Wolf shrinkage improves covariance estimation but does not eliminate this limitation.

Equal Weight is transparent, reproducible and straightforward to monitor. However, equal capital weights do not imply equal risk contributions, and the strategy's success depends on the underlying investment universe.

7.4 Comparison with Risk Parity

The selected SAA is itself the Equal Weight benchmark: with nine ETFs, each target capital weight is 1/9. Risk Parity (equal risk contribution, ERC) provides a complementary diagnostic: it equalizes each ETF's percentage contribution to total portfolio volatility rather than its capital weight. Table 9 compares the two allocations using the same 60-month covariance matrix as the risk budget in Section 9; ERC is evaluated without the 25% individual-weight cap.

Table 9: Equal Weight versus unconstrained Risk Parity (60-month risk window, October 2021–September 2026)

	Equal Weight (SAA)		**Risk Parity (ERC)**	
ETF	**Weight**	**Risk contr.**	**Weight**	**Risk contr.**
IVV	11.11%	14.59%	4.60%	11.11%
VEA	11.11%	16.92%	3.80%	11.11%
XLV	11.11%	12.65%	5.40%	11.11%
XLE	11.11%	23.78%	3.46%	11.11%
TIP	11.11%	5.52%	10.47%	11.11%
SHY	11.11%	1.18%	39.00%	11.11%
IGSB	11.11%	2.53%	20.65%	11.11%
GLD	11.11%	11.28%	5.39%	11.11%
PDBC	11.11%	11.55%	7.24%	11.11%
Ex ante volatility		**8.69%**		**4.70%**

Unconstrained Risk Parity allocates nearly 60% of capital to short-duration bonds and produces an ex ante volatility of 4.70%, well below the required 7–9% range. Achieving the target volatility with this allocation would require leverage, which is prohibited by the investment mandate. Therefore, the unconstrained ERC solution does not satisfy our risk objective, despite achieving more balanced risk contributions.

8 Consistency with the Outlook and Exposures

8.1 Consistency with the Macroeconomic Outlook

Portfolio D combines exposure to economic growth with instruments that may respond differently to inflation, interest-rate and supply-side shocks. The allocation is consistent with the scenarios developed in Section 2, but is not designed to depend on a single macroeconomic forecast.

IVV and VEA provide developed-market equity exposure, while XLV adds defensive healthcare characteristics and XLE increases sensitivity to energy prices. However, energy equities remain exposed to broad equity-market risk and do not represent a direct commodity hedge.

Within fixed income, SHY and IGSB limit nominal interest-rate duration relative to intermediate- and long-maturity bonds. TIP adds inflation-linked Treasury exposure, although its price remains sensitive to changes in real yields. GLD and PDBC provide exposure to gold and diversified commodity futures, respectively; their diversification benefits depend on the type of shock and are not guaranteed.

The main trade-off is that a disinflationary slowdown or a reversal in energy prices could weaken XLE and PDBC, while simultaneous equity losses and rising correlations could reduce the protection offered by diversification.

8.2 Asset-Class Allocation

Portfolio D allocates 44.44% of capital to equities, 33.33% to fixed income and 22.22% to real assets (Figure 3). This classification alone does not describe the distribution of portfolio risk, which is examined in Section 8.4.

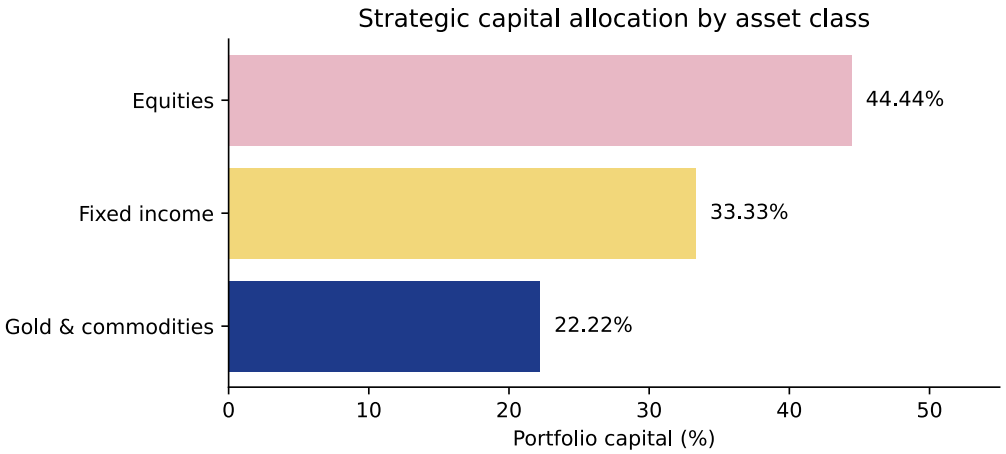


Figure 3: Strategic asset-class weights, Portfolio D.

8.3 Geographic and Sector Exposure

The portfolio has a material US exposure: IVV, XLV and XLE invest in US equities, while TIP, SHY and IGSB primarily provide US fixed-income exposure. VEA adds developed-market equities outside the United States. GLD and PDBC are best classified by their underlying gold and commodity exposures rather than by the country where their ETFs are listed.

Healthcare and energy each account for 11.11% of total capital through XLV and XLE alone. Their *effective* equity-sector exposures are higher because IVV and VEA also contain companies operating in those sectors. Conversely, a precise geographic or sector look-through requires the dated constituent weights of IVV and VEA; ETF-level weights should not be presented as exact underlying country or sector allocations.

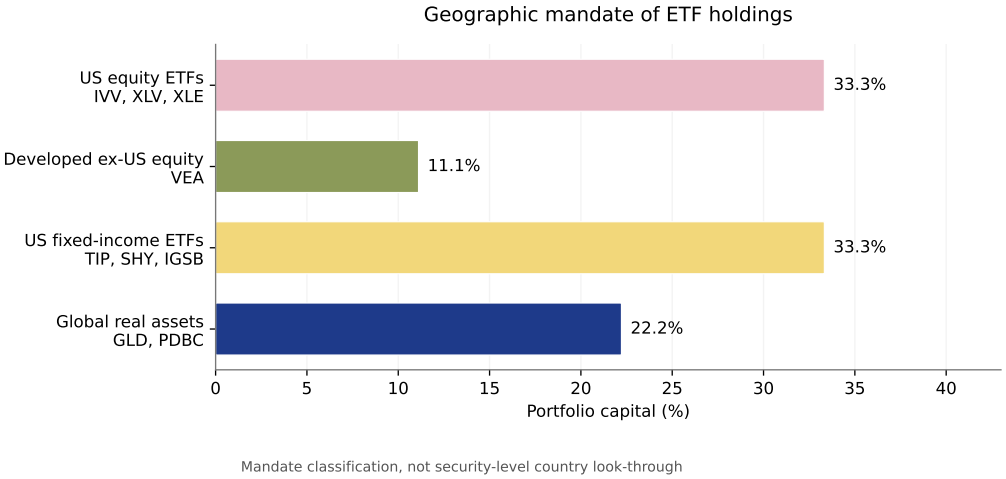


Figure 4: ETF mandate geography; not a security-level country look-through.

8.4 Capital Allocation Versus Risk Allocation

Despite receiving only 44.44% of capital, equities contribute 67.94% of portfolio volatility. Fixed income accounts for 33.33% of capital but just 9.23% of risk, while real assets contribute 22.83% of risk against a 22.22% capital allocation.

The portfolio is therefore equally weighted in capital, but its risk is dominated by equities and especially by energy-sector exposure. The calculation and the individual ETF contributions are presented in Section 9.

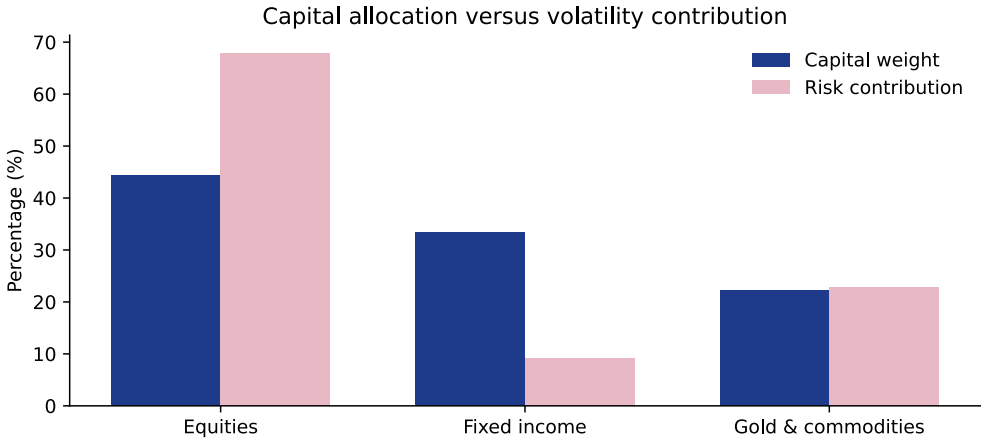


Figure 5: Portfolio D: capital weights versus contributions to total volatility by asset class.

9 Risk Budget and Value at Risk

9.1 Ex Ante Volatility

The risk budget is estimated for Portfolio D – Equal Weight using 60 monthly USD returns from October 2021 to September 2026. This recent risk-estimation window differs from the in-sample period used for portfolio construction. Calculations are documented in the `RISK BUDGET` worksheet of the accompanying workbook.

$$\sigma_{P,\text{ann}} = \sqrt{12\,\mathbf{w}^{\top}\Sigma_m\mathbf{w}} = 8.69\%.$$

The estimated volatility of 8.69% satisfies the 7–9% mandate range, exceeding the central 8% target by 0.69 percentage points. As a model-based estimate, it remains sensitive to changes in market volatilities and correlations.

9.2 Value at Risk

The one-month, 95% parametric Value at Risk (VaR) is estimated using a normal approximation, without a mean-return adjustment:

$$\text{VaR}_{95\%,\,1m} = 1.645\,\frac{\sigma_{P,\text{ann}}}{\sqrt{12}} \approx 4.13\%.$$

For USD 100 million of initial capital, this corresponds to approximately USD 4.13 million. Under the assumed return distribution and fixed risk parameters, the model assigns a 5% probability to a one-month loss exceeding this amount.

Over the same 60-month window, the historical fifth-percentile monthly return was approximately -3.07%, compared with a worst observed monthly return of -5.70%. The parametric VaR threshold of -4.13% lies between these two observations, illustrating the uncertainty surrounding tail-risk estimates.

9.3 Risk Contributions

Equal capital allocation does not imply equal contributions to total portfolio volatility. With Σ_m denoting the monthly covariance matrix, the percentage contribution of asset i to portfolio volatility is

$$\mathrm{RC}_i = \frac{w_i(\Sigma_m \mathbf{w})_i}{\mathbf{w}^\top \Sigma_m \mathbf{w}}, \qquad \sum_i \mathrm{RC}_i = 1.$$

XLE contributes the largest share of portfolio risk (23.78%), despite receiving only 11.11% of capital. Conversely, SHY and IGSB jointly account for 22.22% of capital but contribute just 3.71% of total risk.

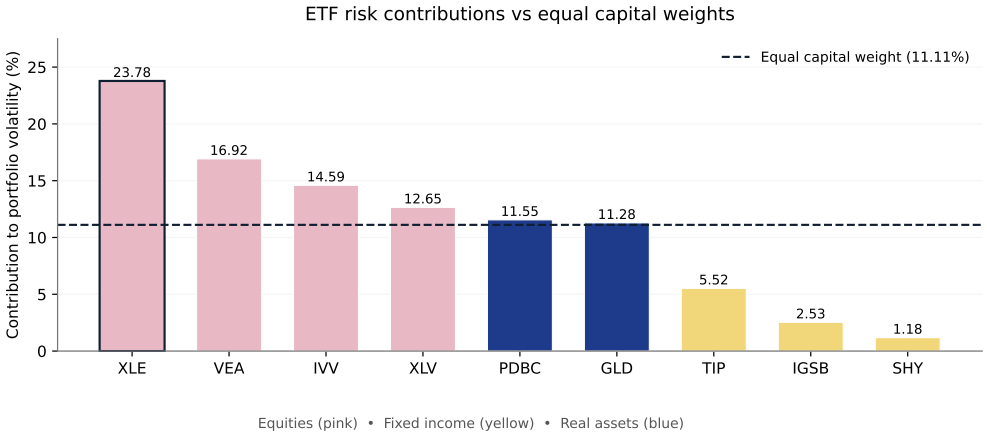


Figure 6: Risk contributions of the nine ETFs compared with the 11.11 percent capital-weight benchmark.

9.4 Drawdown and Risk Management Implications

The maximum historical drawdown calculated over the 60-month risk-estimation window is approximately −10.66%, within the team's internal 15% drawdown threshold. It is larger than the −2.40% maximum drawdown of the 12-month out-of-sample backtest because it covers a longer window that includes the 2022 drawdown. The 15% threshold represents an internal monitoring limit rather than a guaranteed maximum loss.

During the competition, portfolio volatility, drawdown, sector concentration and the changing contributions to risk will be monitored. Any subsequent tactical allocations will be assessed relative to the selected Strategic Asset Allocation and the applicable tracking-error constraints.

10 Conclusion

Quontech selects Portfolio D – Inflation Resilient, Equal Weight as its Strategic Asset Allocation, following a comparison of four investment universes and seven weighting methodologies evaluated through an out-of-sample backtest.

The allocation combines developed-market equities, short-duration and inflation-linked bonds, gold and commodities, consistent with an outlook of uneven growth, persistent inflation risks and monetary-policy uncertainty.

D – Equal Weight achieved the highest observed out-of-sample Sharpe Ratio (1.598), with realized volatility of 8.08% and a maximum drawdown of -2.40%. Its ex ante volatility of

8.69% also satisfies the prescribed 7–9% risk range.

Equal Weight was preferred for its transparency and limited dependence on estimation-sensitive inputs. However, the allocation remains exposed to equity-sector concentration, particularly through XLE.

The selected SAA provides a disciplined strategic benchmark rather than a guarantee of superior performance across future market conditions. Volatility, drawdown and concentration risks will be monitored throughout the competition, with any tactical adjustments evaluated against this benchmark and the applicable tracking-error constraints.

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